Bass Diffusion and Product Life-Cycle Simulation

New-Product Adoption and Product Rollover Timing

Portfolio Project Report

Keman Xiang

kemanxiang005@gmail.com

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Abstract

This report develops a **stochastic Bass diffusion simulation** for a stylized new product, **Q-Phone**, and uses the model to interpret sales decline, identify product life-cycle milestones, and inform product-rollover timing. The simulation represents a finite market of 1,000,000 potential adopters. **Adoption** is governed by an **innovation coefficient p = 0.01** and an **imitation coefficient q = 0.20**, and the market is simulated over 60 monthly periods.

Rather than treating the Bass process only as a deterministic adoption curve, the workbook implements a **stochastic macro simulation** in which the number of remaining prospects evolves through binomial random draws. A single simulated path illustrates the rise-and-fall pattern of monthly sales, while 1,000 replicated life cycles quantify uncertainty in the timing of maturity and decline. Across these replications, the mean peak-sales month is 15.116, the mean decline-stage start is 39.488, the mean mature-stage duration is 24.372 months, and the mean latest product-rollover benchmark is 38.486 months. Convergence diagnostics show that the estimated timing measures stabilize as the number of replications increases. Parameter experiments further distinguish scale effects from diffusion-dynamic effects.

The analysis demonstrates how diffusion theory and Monte Carlo simulation can distinguish a **natural life-cycle decline** from a deterioration in execution and translate adoption dynamics into **product-planning decisions**.

Keywords: Bass diffusion model; Monte Carlo simulation; product life cycle; new-product adoption; product rollover; marketing analytics.

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1. Business Problem and Analytical Objective

Q-Phone is modeled as a newly launched durable product with an estimated addressable market of **one million consumers**. The managerial problem arises when monthly sales begin to decline even though cumulative adoption is still increasing. A naïve interpretation would treat falling monthly sales as evidence of deteriorating marketing performance. A **diffusion perspective** offers a different explanation: **sales can decline endogenously because the pool of remaining prospects becomes progressively smaller after adoption has spread through the market**.

The purpose of the project is therefore not merely to forecast sales. It is to **diagnose where the product sits in its life cycle** and to **convert that diagnosis into an operational timing decision**. Four questions organize the analysis: **(1) what does a simulated 60-month life cycle look like**; **(2) when does the product reach peak sales and enter decline**; **(3) how early must the next generation be launched to avoid waiting until the current product is already in decline**; and **(4) how sensitive are these conclusions to the assumed market size and diffusion parameters?**

2. Theoretical Framework

2.1 Bass Diffusion Logic

The **Bass diffusion model** explains the timing of first purchase through two behavioral forces: **innovation and imitation**. In the original formulation, adoption can occur because consumers possess an intrinsic propensity to adopt independently of prior buyers, or because existing adopters increase the attractiveness or visibility of the product through social influence. This distinction provides a compact behavioral explanation for the characteristic S-shaped cumulative adoption curve and the unimodal sales curve observed in many new-product settings.

In the workbook, m denotes total market potential, Mt the number of prospects remaining at month t, Nt cumulative adopters, Xt monthly sales, and Ft = Nt/m the fraction of the market that has adopted. Conditional adoption is represented by the discrete-time Bass relationship:

$$f_t = (1 - F_t)(\mathbf{p} + \mathbf{q}F_t)$$

Here **p = 0.01** is the **innovation rate** and **q = 0.20** is the **imitation or contagion rate**. Early in the life cycle, adoption is supported by the innovation term. As the installed base grows, the qF_t component strengthens, creating acceleration. Eventually, however, the factor $(1 - F_t)$ becomes small because fewer prospects remain. The same diffusion mechanism that initially accelerates sales can therefore generate a later decline without requiring any deterioration in product quality or managerial execution.

2.2 Stochastic Macro Simulation

The project implements a stochastic, aggregate-market version of the diffusion process. Given Mt prospects and period adoption probability f_t, the next period's remaining prospects are generated from a binomial distribution. The workbook operationalizes this transition with a binomial inverse function driven by a random draw. Cumulative adopters and monthly sales are then recovered from the state transition:

$$M_{t+1} \sim \text{Binomial}(M_t, 1 - f_t)$$

$$N_t = m - M_t$$

$$X_t = N_t - N_{t-1}$$

$$F_t = N_t \,/\, \mathrm{m}$$

This stochastic formulation is important for managerial interpretation. A deterministic diffusion curve would return one timing path. The Monte Carlo formulation instead produces a distribution of possible life cycles under the same structural assumptions, allowing timing decisions to be expressed with sampling uncertainty.

3. Simulation Design and Project Workflow

Component	Workbook specification	Analytical role
Market potential (m)	1,000,000 consumers	Defines the finite adoption pool.
Innovation rate (p)	0.01	Captures adoption independent of prior adopters.
Imitation rate (q)	0.20	Captures social influence / contagion from existing adopters.
Simulation horizon (T)	60 months	Tracks diffusion through later life-cycle stages.
Monte Carlo replications (N)	1,000 life cycles	Quantifies uncertainty in lifecycle timing and rollover decisions.

The project proceeds in **four stages**. First, one **60-month life cycle** is simulated to visualize monthly sales, cumulative adoption, and remaining market potential. Second, the life cycle is **replicated 1,000 times** and **three timing measures** are recorded: **tp**, the month in which sales peak and maturity begins; **td**, the month in which adoption reaches the 80% threshold used to define the start of decline; and **d = td − tp**, the mature-stage duration. Third, **product rollover** is evaluated by treating the month immediately before decline as the latest launch benchmark for the next generation. Fourth, the workbook **varies m, p, and q** to study how scale and diffusion behavior affect the sales path and lifecycle timing.

4. Results

4.1 Representative 60-Month Diffusion Path

In the representative path stored in the workbook, monthly sales rise as the imitation mechanism strengthens, reach a maximum of 34,514 units in **month 15**, and then decline. At the sales peak, cumulative adoption is 331,341 consumers, or **33.13%** of the one-million-consumer market. By **month 40**, cumulative adoption reaches 800,358, crossing the model's **80%** threshold for the decline stage. These values are path-specific realizations rather than population-level timing estimates; their role is to make the diffusion mechanism visible.

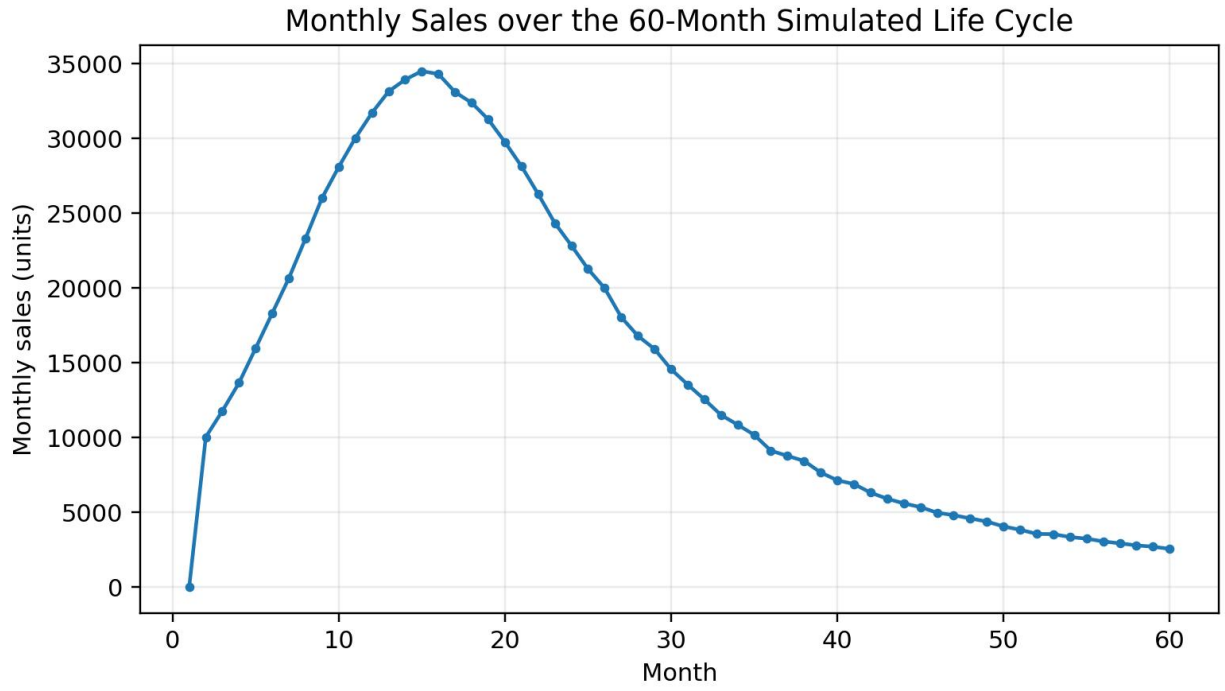


Figure 1. Monthly sales in the representative 60-month simulation path. Source: project Excel workbook.

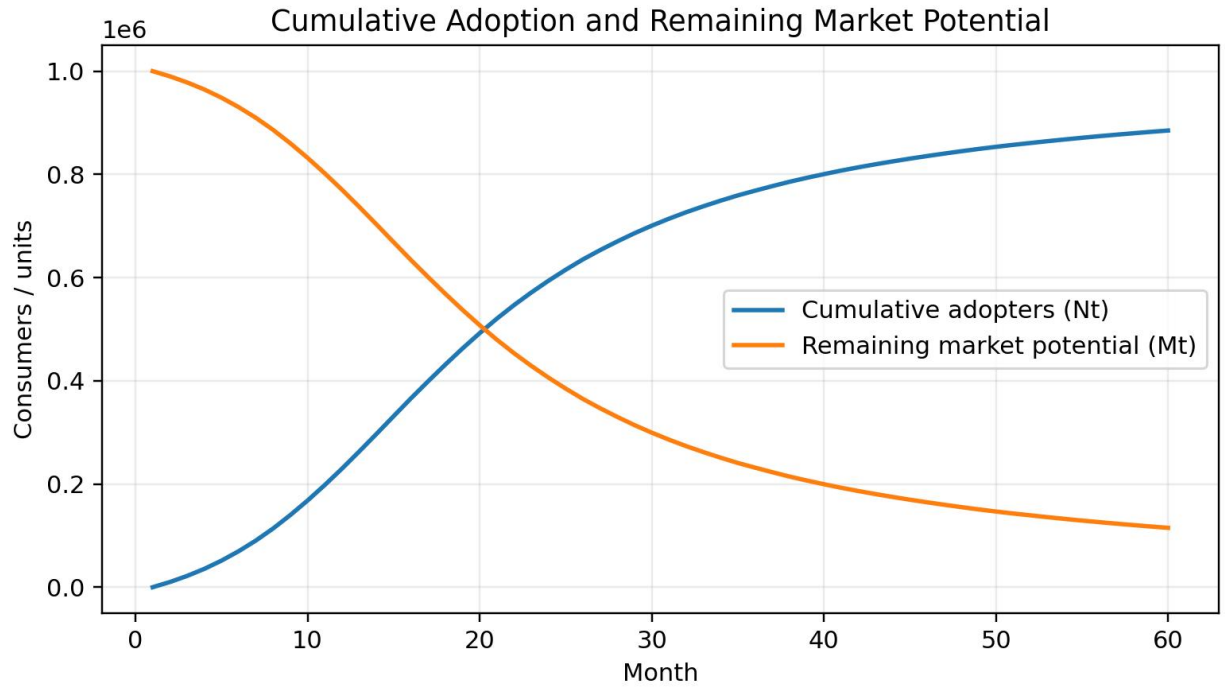


Figure 2. Cumulative adopters and remaining market potential in the representative simulation path. Source: project Excel workbook.

4.2 Monte Carlo Lifecycle Timing

Metric	Mean (months)	95% CI low	95% CI high	Observed range

Peak / maturity start, tp	**15.116**	15.085	15.147	14–16
Decline-stage start, td	**39.488**	39.457	39.519	39–40
Mature-stage duration, d	**24.372**	24.329	24.415	23–26
Latest rollover benchmark	**38.486**	38.455	38.517	38–39

The 1,000-run results are tightly concentrated. **Peak sales** occur on average in **month 15.116** (95% CI for the simulated mean: 15.085–15.147). The **decline threshold** is reached on average in **month 39.488** (95% CI: 39.457–39.519), implying a mean mature-stage **duration of 24.372 months** (95% CI: 24.329–24.415). Across the simulated life cycles, peak timing ranges from months 14 to 16, decline timing from months 39 to 40, and mature-stage duration from 23 to 26 months.

The rollover analysis shifts attention from describing the current product to planning the next one. The **latest launch benchmark** averages **month 38.486**, with a 95% confidence interval of 38.455–38.517 months and an observed simulation range of months 38–39. Under the model's decision rule, the next generation should be ready before the current product enters the decline stage rather than after a prolonged decline becomes visible in realized sales.

4.3 Simulation Convergence

Runs	Mean peak month	Mean decline month	Mean mature duration
25	15.240	39.480	24.240
50	15.100	39.460	24.360
100	15.070	39.500	24.430
250	15.140	39.504	24.364
500	15.124	39.506	24.382
750	15.127	39.489	24.363
1000	15.116	39.488	24.372

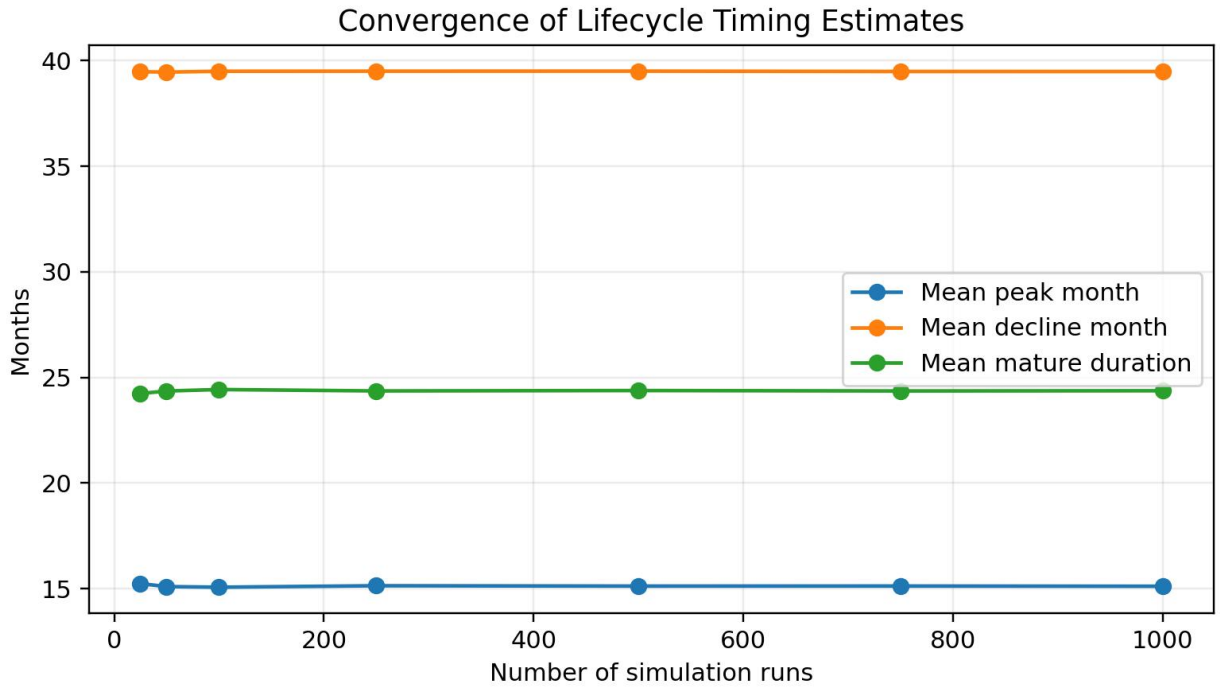


Figure 3. Convergence of lifecycle timing estimates as the number of simulated life cycles increases. Source: project Excel workbook.

The convergence diagnostic evaluates simulation precision rather than changing the diffusion model. With only 25 runs, the estimated peak month is 15.240 and the mature-stage duration is 24.240 months. As the replication count increases, the running estimates fluctuate within progressively narrower bands and settle near the 1,000-run values of 15.116, 39.488, and 24.372 months. The stabilization supports the use of 1,000 replications as a numerically adequate basis for the reported mean timing measures in this workbook.

4.4 Parameter Sensitivity

The parameter experiments separate scale effects from diffusion-speed effects. When market potential **m** is increased from 1 million to 10 million while p and q are held at their baseline values, the workbook shows larger sales volumes but essentially **unchanged lifecycle timing**: mean peak timing remains about 15.116 months and mature-stage duration remains about 24.37 months. This is consistent with m acting primarily as a scale parameter in the simulated system.

Changes in p and q affect the diffusion dynamics more directly. In the workbook's p experiment, increasing **p** from 0.01 to 0.90 leaves mean peak timing close to 15.10 months but gradually shortens the mature-stage duration from 24.372 to 24.347 months. In the q experiment, increasing **q** from 0.20 to 0.90 shifts mean peak timing from 15.116 to 15.108 months; the mature-stage duration remains close to 24.4 months over the tested range. These numerical changes are modest in the workbook's chosen parameter grid, so the appropriate conclusion is not that the parameters are unimportant in general, but that **lifecycle timing is relatively stable over these particular simulated scenarios**.

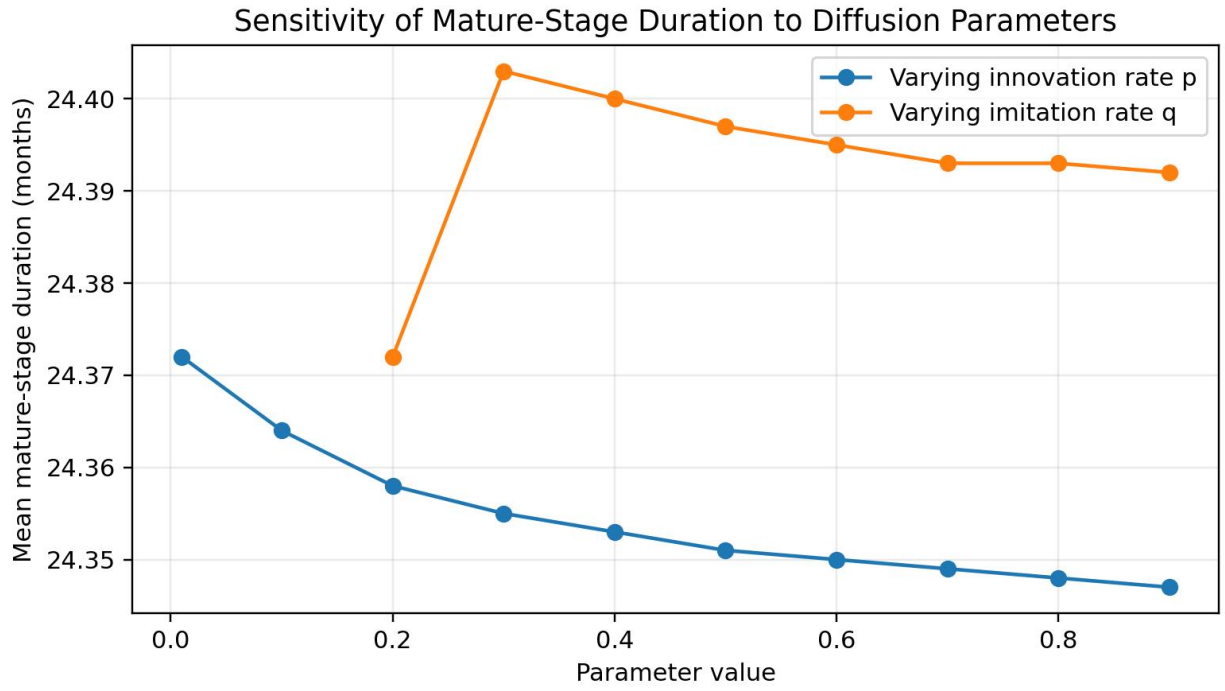


Figure 4. Sensitivity of mature-stage duration to the innovation and imitation parameters in the workbook experiments.

5. Interpretation and Managerial Implications

5.1 Diagnosing Declining Sales

The central managerial insight is that declining monthly sales are not sufficient evidence of poor execution. In the simulated Bass process, sales decline emerges endogenously after diffusion has accumulated a large installed base. Imitation initially accelerates adoption, but the remaining pool of prospects eventually becomes the binding constraint. Consequently, **managers should interpret sales trends jointly with cumulative penetration and remaining market potential**.

5.2 Product-Rollover Planning

The rollover result turns the diffusion model into a planning instrument. The simulated decline threshold occurs around month 39.5, while the latest launch benchmark is approximately month 38.5. Operationally, this means that **development, supply-chain preparation, channel readiness, and launch execution for the successor product must occur well before the current product visibly reaches decline**. The model does not determine the development lead time; instead, it supplies the market-side deadline against which that lead time can be planned.

5.3 Managing with Distributions Rather Than a Single Forecast

A single simulated path is useful for visualization but is a weak basis for a managerial commitment. **Replication** changes the nature of the decision support: peak, decline, and rollover timing become distributions rather than point events. The narrow confidence intervals in this experiment indicate that the Monte Carlo estimate of the mean timing is stable under the stated parameter assumptions. Managers can therefore distinguish two types of uncertainty: random variation generated by the stochastic adoption process,

which is quantified here, and **structural uncertainty** about m, p, q, competitive response, and market conditions, which requires **scenario analysis** or **empirical parameter estimation**.

6. Limitations

The simulation is deliberately parsimonious. Market potential is fixed; each consumer adopts at most once; p and q remain constant within a scenario; and the model does not separately represent price changes, advertising, competitor entry, repeat purchases, capacity constraints, customer heterogeneity, seasonality, or supply disruptions. The 80% adoption threshold is an analytical definition used to mark decline, not a universal empirical rule. The project also treats m, p, and q as specified inputs rather than parameters estimated from observed Q-Phone sales. Accordingly, the results should be interpreted as a structured simulation of diffusion mechanics and timing under stated assumptions, not as a validated real-world forecast.

A further distinction concerns statistical uncertainty. The reported 95% confidence intervals describe uncertainty in the Monte Carlo estimate of the mean lifecycle timing. They should not be interpreted as prediction intervals containing 95% of future product life cycles. Structural uncertainty in the diffusion parameters could be substantially larger than the simulation error shown by the convergence analysis.

7. Conclusion

This project demonstrates how a **stochastic Bass diffusion model** can connect **consumer-adoption theory** with **managerial product planning**. The representative simulation reproduces a **natural life-cycle pattern** in which monthly sales accelerate, peak, and decline while cumulative adoption continues to grow.

Across 1,000 life cycles, maturity begins at approximately **month 15.1**, decline begins at approximately **month 39.5**, and the mature stage lasts about **24.4 months**. The corresponding latest rollover benchmark is approximately **month 38.5**. Convergence analysis indicates that these simulated mean timings are **stable** at the chosen replication count, while parameter experiments distinguish market-size effects from diffusion-dynamic effects.

The broader lesson is that sales decline should be interpreted within the state of **market diffusion**, and that **simulation** can transform that state into **explicit timing information** for product-roadmap decisions.

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Appendix A. Key Workbook Variables

Symbol	Definition	Workbook role
m	Total market potential	Input
p	Innovation rate	Input
q	Imitation / contagion rate	Input
Mt	Remaining prospects at month t	State variable
Xt	Sales in month t	Outcome variable
Nt	Cumulative adopters through month t	State variable
Ft	Fraction of market adopted (Nt/m)	State variable
tp	Peak-sales month / start of maturity	Lifecycle timing output
td	Month adoption reaches the 80% decline threshold	Lifecycle timing output
d	Mature-stage duration (td − tp)	Lifecycle timing output